

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11 – SUB-STRAND 3.3: EXCRETION AND HOMEOSTASIS IN ANIMALS – FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This document is a MODEL to help you answer our driving question: How do different animals maintain a balanced internal environment and eliminate toxic wastes despite drastic changes in their external conditions? Read each section, study the exemplar, then create your own annotated poster of a mammal (for example a goat or a human in Kisumu) showing how it regulates its internal environment and excretes waste. Use your own words and Kenyan examples. Refer back to our anchoring phenomenon: the panting dog, the still lizard on a warm rock, the goats producing concentrated urine, and the freshwater fish producing dilute urine. Label every part of your poster and use arrows and short notes to explain what happens and why. Your poster will be scored with the rubric at the end.

Section 1: Why Animals Must Remove Wastes and Stay Balanced

Explain why all the animals in our phenomenon (dog, lizard, goat, freshwater fish) must get rid of metabolic wastes and keep their internal conditions steady. Define excretion and homeostasis. Use the phenomenon to say what would happen if wastes built up or body temperature and water levels went out of balance.

Every living cell makes waste when it carries out metabolism. When a goat in the Rift Valley digests proteins, its cells break down amino acids and make ammonia, which is very poisonous. Excretion is the removal of metabolic waste products from the body. Homeostasis is keeping the internal environment (temperature, water, salts and blood glucose) within narrow limits even when the outside conditions change. The dog, lizard, goat and fish all stand in the same hot afternoon, but each must keep its cells working. Enzymes only work well within a certain temperature range, and cells swell or shrink if the water balance is wrong. If toxic wastes build up, cells are damaged and the animal can die. So all four animals need excretion and homeostasis, but they use different methods that suit their bodies and habitats.

Section 2: Different Nitrogenous Wastes for Different Habitats

Compare how different animals get rid of nitrogenous waste (ammonia, urea, uric acid).

Animals deal with toxic ammonia in different ways depending on their habitat. Fish living in Lake Victoria have plenty of water around them, so they can release ammonia directly across their gills, or make some urea, and lose it in very dilute urine. A freshwater fish also takes in water by osmosis all the time, so its kidneys must remove the extra water, which is why it excretes

Explain why the freshwater fish and the goat produce such different urine, and link the type of waste to how much water the animal has available.	large amounts of dilute urine. Mammals like goats and humans cannot afford to lose that much water on land, so the liver changes ammonia into urea, which is less toxic. The kidneys then remove urea in urine that can be concentrated to save water. Insects and birds go even further: they change ammonia into uric acid, which is almost insoluble and leaves as a paste, using very little water. This shows that the type of waste matches how much water is available to the animal.
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Section 3: How the Mammalian Excretory Organs Work

Describe the roles of the liver, kidney (nephron), bladder and skin in excretion in a mammal. Explain ultrafiltration and selective reabsorption and trace the path of urine. Include these as labelled parts on your poster.	In a mammal, the liver breaks down excess amino acids in a process called deamination and makes urea. It also processes other toxins. Urea travels in the blood to the kidneys. Each kidney has thousands of nephrons. In the glomerulus, high blood pressure pushes water, urea, glucose, salts and small molecules out of the blood into the Bowman's capsule. This is ultrafiltration, and large molecules such as proteins and blood cells stay in the blood. As the filtrate moves along the tubule, useful substances such as all the glucose, most of the water and some salts are taken back into the blood. This is selective reabsorption. What remains, water, urea and extra salts, is urine. Urine flows from the kidney through the pelvis and ureter to the bladder, where it is stored until it leaves the body. The skin also excretes: when you sweat after playing football in Nairobi, sweat carries water, salts and a little urea out of the body.
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Section 4: Negative Feedback Keeps the Body Stable

Explain how negative feedback loops regulate body temperature, water balance (osmoregulation using ADH) and blood glucose (insulin and glucagon). Use the panting dog, the basking lizard and the goat to show different responses to the same hot conditions.	Negative feedback means that when a condition changes away from normal, the body responds to bring it back. For temperature, when a dog gets too hot, receptors tell the brain, and the dog pants so that water evaporates from its tongue and cools it. In humans, sweat glands release sweat and blood vessels in the skin widen (vasodilation) so that more heat is lost. A lizard is not able to cool itself in the same way, so it uses behaviour: it basks on a warm rock to warm up and moves to shade when it is too hot. For water balance, if a goat is short of water, the blood becomes more concentrated. The brain detects this and the pituitary gland releases ADH, which makes the kidney tubules reabsorb more water, so the urine is small in volume and concentrated. When the goat drinks plenty, less ADH is released and the urine becomes more dilute. For blood glucose, after eating ugali, glucose rises and the pancreas releases insulin, so cells and the liver take up glucose. When glucose falls, the pancreas releases glucagon, which makes the liver release glucose. Each loop reverses the change and keeps conditions steady.
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Section 5: Putting It All Together – Our Final Explanation

Write a short final explanation answering the driving question. Use your annotated mammal poster as evidence. Connect excretion, homeostasis and the differences between the dog, lizard, goat and fish. Finish by saying why these systems are essential for survival.	Animals survive changing conditions because they have excretory and homeostatic systems adapted to their environments. The dog pants and the lizard basks because they use different ways to control body temperature. The goat conserves water by making concentrated urine under the control of ADH, while the freshwater fish gets rid of excess water in dilute urine. In mammals, the liver turns toxic ammonia into urea, the kidney nephrons filter blood by ultrafiltration and reabsorb useful substances to form urine, and the skin loses some water, salts and urea in sweat. Negative feedback loops using sweating, vasodilation, behaviour, ADH, insulin and glucagon keep temperature, water and glucose steady. Together these systems stop toxic wastes from building up and protect cells from damage, so animals can survive in different environments from the hot Rift Valley to Lake Victoria.
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific accuracy of excretory processes	Correctly and fully explains deamination, urea formation, ultrafiltration, selective reabsorption and the urine pathway, with all organs labelled correctly and the roles of liver, kidney, bladder and skin clearly described.	Describes most excretory processes and organs correctly, with only minor errors or missing details.	Names some organs but has major errors or gaps in explaining how wastes are formed, filtered or removed.
Explanation of homeostasis and negative feedback	Clearly explains negative feedback for temperature, water balance (ADH) and blood glucose (insulin and glucagon), showing stimulus, receptor, response and return to normal.	Explains at least two feedback loops correctly, though some steps of the loop are missing.	Mentions homeostasis but does not show how feedback works, or confuses the hormones and responses.
Connection to the phenomenon and comparison of animals	Uses the dog, lizard, goat and freshwater fish to explain how habitat shapes waste type, water balance and temperature control, with clear Kenyan examples.	Refers to the phenomenon animals and makes some correct comparisons, but links to habitat are limited.	Barely refers to the phenomenon or makes incorrect or no comparisons between the animals.
Poster design, annotation and labelling	Poster is neat, well organised and fully labelled, with arrows and annotations that show how the parts work together and cause and effect clearly.	Poster has most parts labelled and some annotations, but the flow or detail is uneven.	Poster has few labels or annotations and is hard to follow.
Use of scientific vocabulary and reasoning	Uses terms such as excretion, homeostasis, osmoregulation, nephron and ADH accurately, and gives clear reasons for why each system is needed for survival.	Uses most scientific terms correctly and gives some reasoning.	Uses few scientific terms or uses them incorrectly, with little reasoning.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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